

Assignment 5

08022311 陈鲲龙

2025 年 3 月 23 日

Problem 1 Description

As a process engineer, you decide to develop a first-order + time delay model of a process using a step test. The process is initially at steady state, with an input flow rate of 5 gpm and an output of 0.75 mol/L. You make a step increase of 0.5 gpm at 3:00 p.m. and do not observe any changes until 3:07 p.m. At 3:20 p.m., the value of the output is 0.8 mol/L. You become distracted and do not have a chance to look at the output variable again, until you leave for happy hour at a local watering hole at 6:30 p.m. You note that the output has ceased to change and has achieved a new steady-state value of 0.85 mol/L. What are the values of the process parameters, with units? Show your work.

Step Test Parameter Calculation

Assuming the standard form of a first-order plus dead time (FOPDT) model is:

$$y(t) = y_0 + K \Delta u \left(1 - \exp \left(-\frac{t - \theta}{\tau} \right) \right)$$

where:

- $y(t)$ is the output at time t ,
- y_0 is the initial steady state output,
- Δu is the input change,
- K is the process gain,
- θ is the dead time (delay time),
- τ is the time constant.

1. Calculation of Process Gain K

Given:

- Initial steady state output: $y_0 = 0.75$ mol/L,
- New steady state output: $y_{ss} = 0.85$ mol/L,

- Step input change: $\Delta u = 0.5$ gpm.

The change in output is:

$$\Delta y = y_{ss} - y_0 = 0.85 - 0.75 = 0.10 \text{ mol/L}$$

Thus, the process gain K is:

$$K = \frac{\Delta y}{\Delta u} = \frac{0.10 \text{ mol/L}}{0.5 \text{ gpm}} = 0.20 \frac{\text{mol/L}}{\text{gpm}}$$

2. Determination of Dead Time θ

It is stated that the step is applied at 3:00 PM, but no change is observed until 3:07 PM. Therefore, we determine:

$$\theta = 7 \text{ minutes}$$

3. Calculation of the Time Constant τ

The response after the dead time is given by:

$$y(t) = y_0 + K \Delta u \left(1 - \exp \left(-\frac{t - \theta}{\tau} \right) \right)$$

At 3:20 PM, the output is $y(3 : 20) = 0.80$ mol/L. Since the response starts at 3:07 PM, the effective response time is:

$$t - \theta = 3 : 20 - 3 : 07 = 13 \text{ minutes}$$

Substituting the data:

$$0.80 = 0.75 + 0.10 \left(1 - \exp \left(-\frac{13}{\tau} \right) \right)$$

Rearranging:

$$0.05 = 0.10 \left(1 - \exp \left(-\frac{13}{\tau} \right) \right)$$

Thus:

$$1 - \exp \left(-\frac{13}{\tau} \right) = 0.5$$

Solving:

$$\exp \left(-\frac{13}{\tau} \right) = 0.5$$

Taking the natural logarithm:

$$-\frac{13}{\tau} = \ln(0.5) \approx -0.693$$

Therefore:

$$\tau = \frac{13}{0.693} \approx 18.8 \text{ minutes}$$

Final Results

- Process Gain: $K = 0.20 \frac{\text{mol/L}}{\text{gpm}}$
- Dead Time: $\theta = 7 \text{ minutes}$
- Time Constant: $\tau \approx 18.8 \text{ minutes}$

$$y(t_{(\min)})_{(\text{mol/L})} = 0.75_{(\text{mol/L})} + 0.2_{(\frac{\text{mol/L}}{\text{gpm}})} \Delta u_{(\text{gpm})} \left(1 - \exp \left(-\frac{t_{(\min)} - 7_{(\min)}}{18.8_{(\min)}} \right) \right)$$

Problem 2 Description

Consider a unit step input change made at $k = 0$, resulting in the output response

k	-1	0	1	2	3	4	5	6	7	8	9	10
y	0	0	0.1044	0.3403	0.6105	0.8494	1.0234	1.1244	1.1616	1.1531	1.118	1.0742
u	0	1	1	1	1	1	1	1	1	1	1	1
k	11	12	13	14	15	16	17	18	19	20		
y	1.0336	1.0023	0.9828	0.9744	0.9742	0.9790	0.9860	0.9929	0.9985	1.0022		
u	1	1	1	1	1	1	1	1	1	1		

Estimate the parameters for a discrete linear model with the form

$$g(z) = \frac{b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

Solution 1

Correspond to the difference equation

$$y[k] + a_1 y[k-1] + a_2 y[k-2] = b_1 u[k-1] + b_2 u[k-2].$$

For a unit step input,

$$u[k] = \begin{cases} 0, & k < 0, \\ 1, & k \geq 0, \end{cases}$$

so that for $k \geq 2$ the equation becomes

$$y[k] + a_1 y[k-1] + a_2 y[k-2] = b_1 + b_2.$$

The DC gain of the system is

$$G = g(1) = \frac{b_1 + b_2}{1 + a_1 + a_2}.$$

Since the step response settles at 1 (i.e. $y[k] \rightarrow 1$), we have

$$\frac{b_1 + b_2}{1 + a_1 + a_2} = 1 \implies b_1 + b_2 = 1 + a_1 + a_2.$$

Step-by-Step Estimation

1. At $k = 1$:

Given $y[1] = 0.1044$ and with initial conditions $y[0] = 0$ and $y[-1] = 0$, the difference equation at $k = 1$ is

$$y[1] + a_1 y[0] + a_2 y[-1] = b_1 u[0] + b_2 u[-1].$$

Since $u[0] = 1$ and $u[-1] = 0$, we obtain

$$0.1044 = b_1,$$

thus

$$b_1 = 0.1044.$$

2. At $k = 2$:

With $y[2] = 0.3403$, $y[1] = 0.1044$, and $y[0] = 0$, we have

$$0.3403 + a_1(0.1044) + a_2(0) = b_1 + b_2.$$

Thus,

$$b_1 + b_2 = 0.3403 + 0.1044 a_1.$$

But from the DC gain condition,

$$b_1 + b_2 = 1 + a_1 + a_2.$$

Equate the two expressions:

$$1 + a_1 + a_2 = 0.3403 + 0.1044 a_1.$$

Rearranging gives:

$$a_2 = -0.6597 - 0.8956 a_1. \quad (1)$$

3. At $k = 3$:

Given $y[3] = 0.6105$, $y[2] = 0.3403$, and $y[1] = 0.1044$, the equation is:

$$0.6105 + a_1(0.3403) + a_2(0.1044) = b_1 + b_2.$$

Using the DC gain condition $b_1 + b_2 = 1 + a_1 + a_2$, we have:

$$1 + a_1 + a_2 = 0.6105 + 0.3403 a_1 + 0.1044 a_2.$$

Rearrange this to obtain:

$$1 - 0.6105 + a_1(1 - 0.3403) + a_2(1 - 0.1044) = 0,$$

or

$$0.3895 + 0.6597 a_1 + 0.8956 a_2 = 0. \quad (2)$$

Substitute the expression for a_2 from (1) into (2):

$$0.3895 + 0.6597 a_1 + 0.8956 \left(-0.6597 - 0.8956 a_1 \right) = 0.$$

This expands to:

$$0.3895 + 0.6597 a_1 - 0.8956 \times 0.6597 - 0.8956^2 a_1 = 0.$$

Numerically, $0.8956 \times 0.6597 \approx 0.5909$ and $0.8956^2 \approx 0.8021$.

Thus:

$$0.3895 + 0.6597 a_1 - 0.5909 - 0.8021 a_1 = 0,$$

or

$$-0.2014 - 0.1424 a_1 = 0.$$

Solving for a_1 gives:

$$a_1 \approx -\frac{0.2014}{0.1424} \approx -1.414.$$

Substitute a_1 back into (1):

$$a_2 = -0.6597 - 0.8956(-1.414) \approx -0.6597 + 1.265 \approx 0.6053.$$

4. Determining b_2 :

From the DC gain condition,

$$b_1 + b_2 = 1 + a_1 + a_2.$$

Substitute the values:

$$1 + a_1 + a_2 = 1 - 1.414 + 0.6053 \approx 0.1913.$$

Since $b_1 = 0.1044$, we have:

$$b_2 = 0.1913 - 0.1044 \approx 0.0869.$$

Final Estimated Parameters

$$a_1 \approx -1.414,$$

$$a_2 \approx 0.6053,$$

$$b_1 \approx 0.1044,$$

$$b_2 \approx 0.0869.$$

Thus, the estimated discrete-time transfer function is:

$$g(z) = \frac{0.1044 z^{-1} + 0.0869 z^{-2}}{1 - 1.414 z^{-1} + 0.6053 z^{-2}}.$$

Solution 2

$$Y = \begin{bmatrix} y(1) \\ \vdots \\ y(N) \end{bmatrix}, \quad \Phi = \begin{bmatrix} y(0) & y(-1) & u(0) & u(-1) \\ \vdots & \vdots & \vdots & \vdots \\ y(N-1) & y(N-2) & u(N-1) & u(N-2) \end{bmatrix},$$

$$\Theta = \begin{bmatrix} -a_1 & -a_2 & b_1 & b_2 \end{bmatrix}^T = (\Phi^T \Phi)^{-1} \Phi^T Y$$

MATLAB code:

```

clear; close all; clc;

%% 1. Measured Data
k_meas = -1:20; % 22 points
y_meas = [0, 0, 0.1044, 0.3403, 0.6105, 0.8494, 1.0234, 1.1244, 1.1616, ...
          1.1531, 1.118, 1.0742, 1.0336, 1.0023, 0.9828, 0.9744, ...
          0.9742, 0.9790, 0.9860, 0.9929, 0.9985, 1.0022];

%% 2. Estimated Model Parameters
% Given estimated discrete-time model:
%  $g(z) = (b_1 z^{-1} + b_2 z^{-2}) / (1 + a_1 z^{-1} + a_2 z^{-2})$ 
Y=zeros(20,4);
Y(:,1)=y_meas(:,3:22)';
Phi(:,1)=y_meas(:,2:21)';
Phi(:,2)=y_meas(:,1:20)';
Phi(:,3)=ones(20,1);
Phi(:,4)=ones(20,1);
Phi(1,4)=0;
Theta=(inv(Phi'*Phi))*Phi'*Y;

a1 = -Theta(1,1);
a2 = -Theta(2,1);
b1 = Theta(3,1);
b2 = Theta(4,1);

```

Final Estimated Parameters

$$a_1 \approx -1.413716515869722,$$

$$a_2 \approx 0.606453686414598,$$

$$b_1 \approx 0.1044000000000000,$$

$$b_2 \approx 0.088330523513179.$$

Thus, the estimated discrete-time transfer function is:

$$g(z) = \frac{0.1044 z^{-1} + 0.0883 z^{-2}}{1 - 1.4137 z^{-1} + 0.6065 z^{-2}}.$$

Comparison:

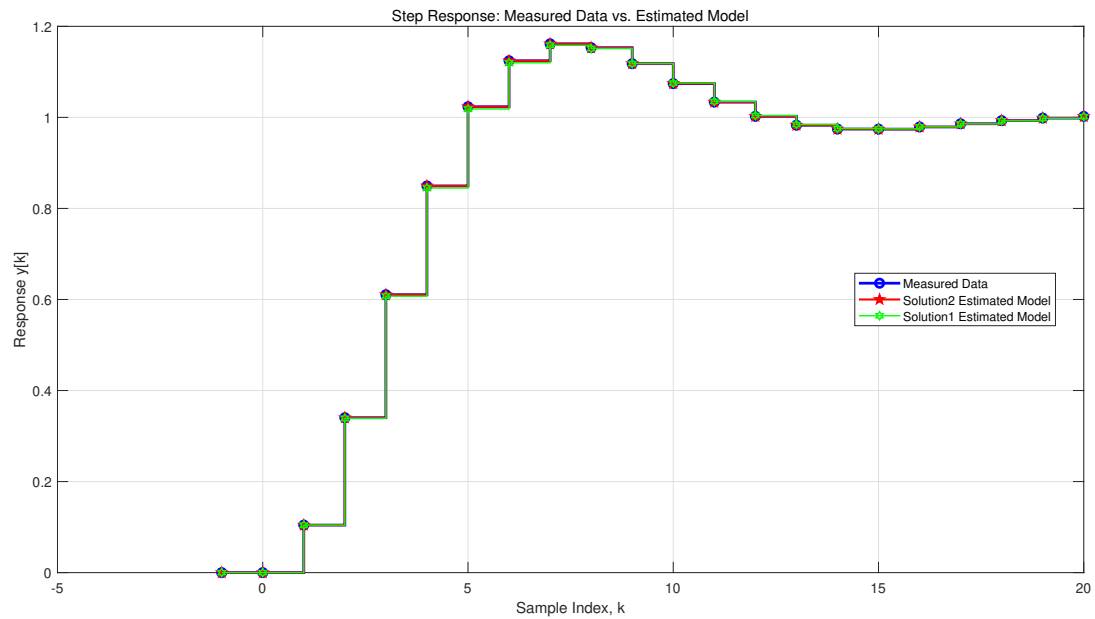


Fig. 1: Solution 1 vs Solution 2

Appendix

MATLAB code:

```
clear; close all; clc

%% 1. Measured Data
k_meas = -1:20; % 22 points
y_meas = [0, 0, 0.1044, 0.3403, 0.6105, 0.8494, 1.0234, 1.1244, 1.1616, ...
          1.1531, 1.118, 1.0742, 1.0336, 1.0023, 0.9828, 0.9744, ...
          0.9742, 0.9790, 0.9860, 0.9929, 0.9985, 1.0022];

%% 2. Estimated Model Parameters
% Given estimated discrete-time model:
% g(z) = (b1*z^-1 + b2*z^-2) / (1 + a1*z^-1 + a2*z^-2)
Y=zeros(20,4);
Y(:,1)=y_meas(:,3:22)';
Phi(:,1)=y_meas(:,2:21)';
Phi(:,2)=y_meas(:,1:20)';
Phi(:,3)=ones(20,1);
Phi(:,4)=ones(20,1);
Phi(1,4)=0;
Theta=(inv(Phi'*Phi))*Phi'*Y;

%% solution2
```

```

a1_solution2 = -Theta(1,1);
a2_solution2 = -Theta(2,1);
b1_solution2 = Theta(3,1);
b2_solution2 = Theta(4,1);

num_solution2 = [0, b1_solution2, b2_solution2]; %  $0z^0 + b1z^{-1} + b2z^{-2}$ 
den_solution2 = [1, a1_solution2, a2_solution2]; %  $1 + a1z^{-1} + a2z^{-2}$ 
Ts = 1; % Sampling time (assumed to be 1 time unit)
sys_solution2 = tf(num_solution2, den_solution2, Ts);
%% 3. Simulate the Step Response from the Estimated Model
numSamples = 20;
[t_sim_solution2, y_sim] = step(sys_solution2, numSamples);
y_est_solution2 = [0; t_sim_solution2];
k_est = -1:20;

%% solution1
a1_solution1 = -1.414;
a2_solution1 = 0.6053;
b1_solution1 = 0.1044;
b2_solution1 = 0.0869;
num_solution1 = [0, b1_solution1, b2_solution1]; %  $0z^0 + b1z^{-1} + b2z^{-2}$ 
den_solution1 = [1, a1_solution1, a2_solution1]; %  $1 + a1z^{-1} + a2z^{-2}$ 
Ts = 1; % Sampling time (assumed to be 1 time unit)
sys_solution1 = tf(num_solution1, den_solution1, Ts);
%% 3. Simulate the Step Response from the Estimated Model
numSamples = 20;
[t_sim_solution1, y_sim] = step(sys_solution1, numSamples);
y_est_solution1 = [0; t_sim_solution1];
k_est = -1:20;
%% 4. Plot the Measured and Estimated Step Responses
figure;
stairs(k_meas, y_meas, 'bo-', 'LineWidth', 2, 'DisplayName', 'Measured Data'); hold on;
stairs(k_est, y_est_solution2, 'rpentagram-', 'LineWidth', 1.5, 'DisplayName',
'Solution2 Estimated Model'); hold on;
stairs(k_est, y_est_solution1, 'ghexagram-', 'LineWidth', 1, 'DisplayName',
'Solution1 Estimated Model');
xlabel('Sample Index, k');
ylabel('Response y[k]');
title('Step Response: Measured Data vs. Estimated Model');
legend('Location', 'Best'); grid on;

```